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My Health @ EU
eHealth Digital Service Infrastructure
Electronic cross-border health services in the EU

OpenNCP Implementation

DOCUMENT VERSION 10.0.0

DATE 01/06/2026

STATUS Wave 10 Operation Ready

DG SANTE, CEF eHealth DSI, 2026
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COVER AND CONTROL PAGE OF DOCUMENT	
Document name:	OpenNCP Implementation
Distribution level*:	PU
Status:	Wave 10 Operation Ready
Author(s): Organization:	eHealth DSI provider

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ABSTRACT
The "Open Source NCP Design" represents the basic document to identify the NCP "core" components, together with their interfaces, that will be centrally developed as Open Source and made available to all through the identified Open Source Community. Country specific components will be also outlined, to allow PN's to adopt their own implementation strategy and the integration strategy with Core components.

CHANGE HISTORY				
Version	Date	Status Changes	From	Review
V10.0.0	01/06/2026	Wave 10 Operation Ready: CP089-EUVAC services & CP090-EUDI Wallet integration updates	eHealth DSI Solution Provider	
V10.0.0-RC	29/04/2026	Wave 10 Release Candidate: CP089-EUVAC services & CP090-EUDI Wallet integration updates	eHealth DSI Solution Provider	
V8.0.1	24/02/2025	Wave 8 Operation Ready: Wave 6 Hotfix Operation Ready CP-077 - Implement Support for Patient Access to translated health data Alignment with v6.5.0	eHealth DSI Solution Provider	
V8.0.1	10/01/2025	Wave 8 Release Candidate: Wave 6 Hotfix Operation Ready CP-077 - Implement Support for Patient Access to translated health data Alignment with v6.5.0	eHealth DSI Solution Provider	
V8.0.0-RC	12/03/2024	Wave 8 Release Candidate – Integration of CP-073 - Implement the new service Laboratory Result Report: FHIR implemetation	eHealth DSI Solution Provider	
V6.5.0	03/11/2023	Wave 6 Hotfix Operation Ready CP-077 - Implement Support for Patient Access to translated health data	eHealth DSI Solution Provider	

		Additional functionality		
V6.5.0-RC	18/10/2023	Wave 6 Hotfix Release Candidate. Implementation of CP-077 - Implement Support for Patient Access to translated health data	eHealth DSI Solution Provider	
V6.4.0	22/08/2023	Wave 6 Hotfix Operation Ready	eHealth DSI Solution Provider	
V6.4.0-RC	26/07/2023	Wave 6 Hotfix Release Candidate	eHealth DSI Solution Provider	CP-077 IMPLEMENT SUPPORT FOR PATIENT ACCESS TO TRANSLATED HEALTH DATA
V6.1.0	14/07/2022	Wave 6 Operation Ready	eHealth DSI Solution Provider	
V6.0.0	29/04/2022	Wave 6 Release Candidate	eHealth DSI Solution Provider	CP-059 Abuse Detection System
V5.0.0	30/04/2021	Wave 5 Release Candidate	eHealth DSI Solution Provider	CP-047 Integration – Submitted for eHMSEG Adoption
V4.0.0	15/05/2020	Wave 4 Operation Ready	eHealth DSI Solution Provider	Administrative Update
V3.0.0	02/07/2019		eHealth DSI Solution Provider	Administrative update (Version-bump). No content updates.
V2.2.0	30/06/2018	Wave 2 Operation Ready	eHealth DSI Solution Provider	
V2.1.0	01/06/2017	Wave 1 Operation Ready Released for eHMSEG adoption	eHealth DSI Solution Provider	
V2.0.0	28/03/2017		eHealth DSI provider	Remove all references to epSOS and requirements
V1.1-RC0	22/05/2014	Final	ELGA	Marcello Melgara

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1 NCP Design Introduction

1.1 Scope

This document specifies the high-level structure of the National Contact Point derived from the System Architecture and the Technical Specifications.
The goal of the activity is to design and develop the OpenNCP as a set of Open Source Components.

1.2 Introduction

The components developed within the OpenNCP MUST adhere to the protocols, standards and profiles set forth by the eHealth DSI Specifications.

The OpenNCP National Infrastructure Interfaces will be defined to operate in close conjunction with the components developed in the eHealth DSI and will not necessarily follow any predefined implementations.
The implementation of the OpenNCP components will only use open source components as third party contributions. Furthermore, the eHealth DSI will strive to develop all components according to existing enterprise standards.
In the implementation the reuse of currently available APIs and components is generally preferred over the implementation of new specific components.
The design of the components must be flexible enough to reasonably easily accommodate the addition of new eHealth DSI services.

2 OpenNCP Specification

2.1 Scope

The contents of the open source NCP common components implementation – OpenNCP - revolves around providing the components that comprise the common NCP infrastructure and which are independent of national infrastructures. Prospectively, the phrase "common components" or "NCP components" refer to the above stated understanding of NCP components. As such, the implementation of the common components is scoped by, in one end: the IHE profiles and in the other end: the integration points to the local infrastructure. The NCP follows the general SOA paradigm, and each component in the NCP must operate as a service which is externally scoped by the NCP, but internally available to the other components. While the NCP as a whole is mandated to operate in a synchronous manner, the individual services that the components implement may operate asynchronously. For instance, the issuing of instructions to log audit events should not halt the processing of the message, only guarantee for its own completion.

2.2 Introduction

Because the NCP is stateless and operates in a synchronous manner, the NCP as a whole can be seen as handling messages between systems and interaction between NCP's revolves around the exchange of messages in a request-reply, point-to-point pattern. The requests are queries from countries that have assumed responsibility for treating a patient, called Country-B and operating via a NCP-B server. The responses are given at the country of affiliation, called Country-A, which operates via a NCP-A server. The high-level view of this setup can be seen in figure 1.



Figure 1: Overview of NCP communication

The NCP fundamentally serves two modes of operation: one as NCP-A and one as NCP-B. Both roles are mediators of messages, where NCP-B acts as an initiating mediator and NCP-A as a receiving mediator.
The common components support all mediating operations within the NCP when it's acting as a NCP-A as well as when it's acting as a NCP-B. The eHealth DSI specification identifies five core services that the NCP servers must support:

1. Identification Service
2. Patient Service
3. Order Service
4. Dispensation Service

5. Original Clinical Document (OrCD) Service
6. Lab Result Report
7. Hospital Discharge Report
8. Medical Imaging
9. FHIR-based Patient Identification

In this specification, these services are considered the business logic supporting the use cases. These services are exposed by the implementation of the IHE profiles XCPD, XCA and XDR. The retrieval of the requested information/setting of information within the national infrastructure is not within the scope of the OpenNCP implementation.
In addition to the above services, this document will specify and describe the NCP components needed to support the internal operations.

Not all of the above listed components are subject for implementation in the OpenNCP implementation. Some of them are third party open source products while other are to be implemented by the PN's. The following sections will describe this in more detail.

2.3 NCP Common Components Architecture

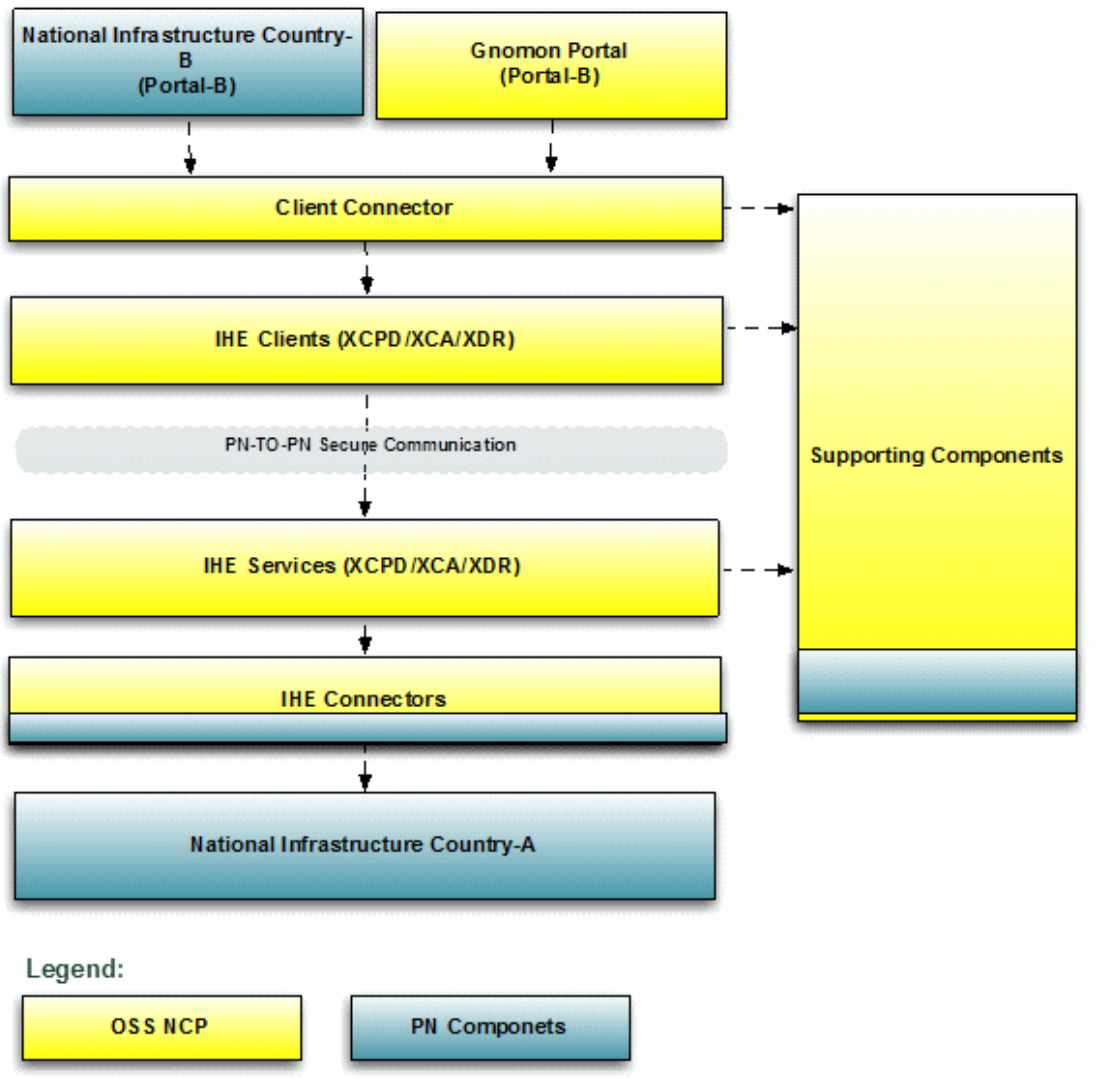


Figure 2: NCP architecture, Logical High Level Model

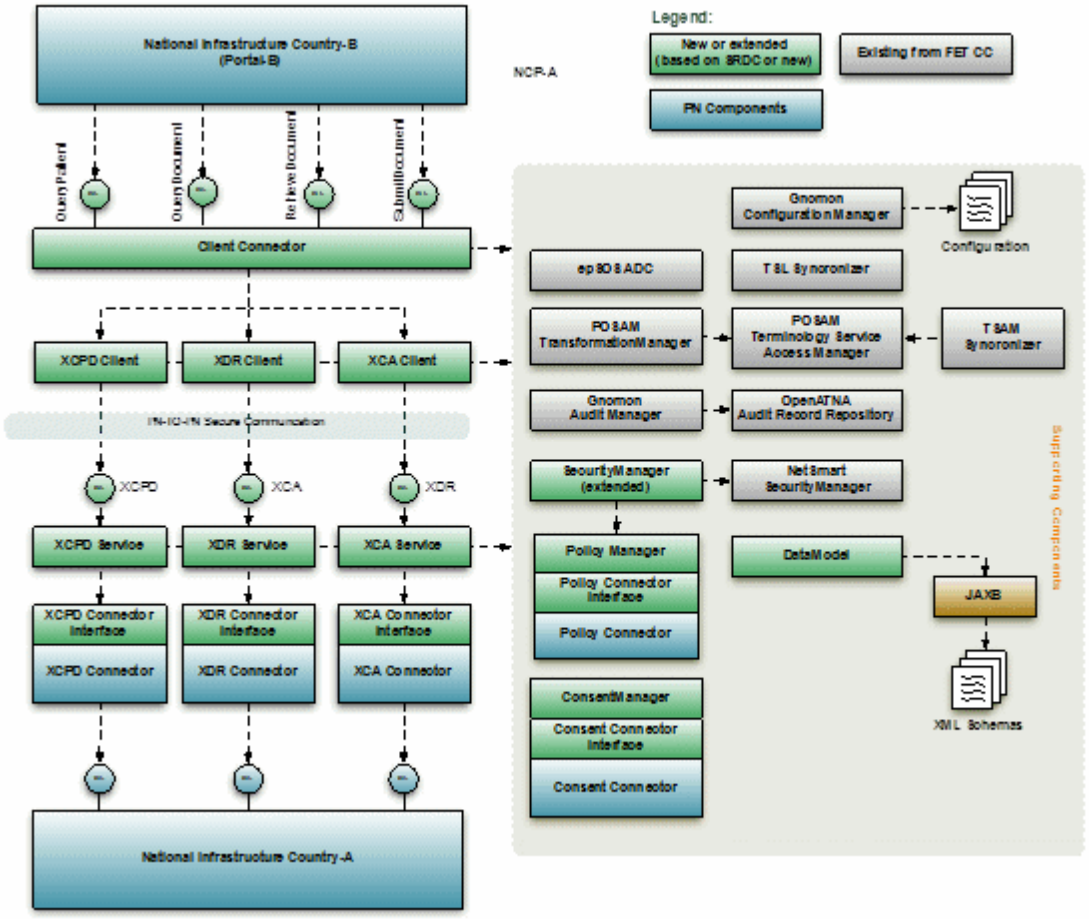


Figure 3: NCP architecture, Logical Component Model

2.4 Components development approach

The development and distribution of the common components are divided into separate software modules, which are described in detail in this chapter.

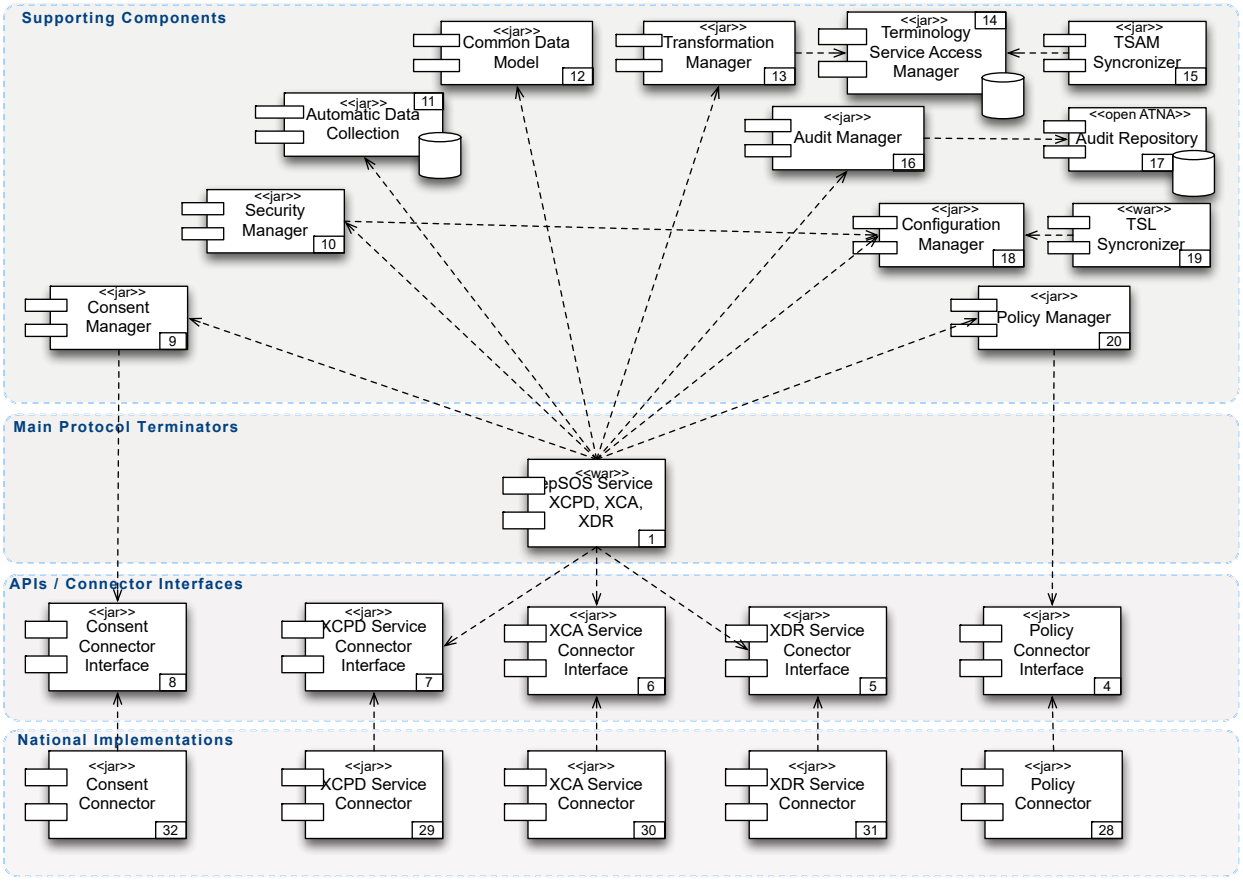


Figure 4: NCP architecture describing deployable components and the dependancies between them

2.4.1 NCP-IHE Service implementations

The preferred approach is the adoption of the SRDC NCP IHE Service implementations (XCPD, XCA, XDR services). The communication with the national infrastructure in the SRDC NCP solution relies on a simple API used for calling nationally implemented services (patient search, patient summary search, patient summary retrieval and consent manager). The workflow management in NCP-A is performed by IHE services.

2.4.1.1 eHealth DSI Services (Protocol Terminators)

The initial implementation of the common components aims at providing only eHealth DSI services. The following sections describe the eHealth DSI services and their scope within the OpenNCP implementation. The operations of the eHealth DSI services are implemented through IHE services within the NCP framework.

2.4.1.2 Order Service

The ordering of prescriptions uses the XCA service to communicate with the National Connector Interface implementations for retrieval of prescriptions. The Order Service needs a National Connector Interface to retrieve the relevant data and, given the presence of this data, uses the business logic within the TransformationManager and the XDR service to deliver dispensation information about the prescription to the requestor.

2.4.1.3 Dispensation Service

The Dispensation Service handles the registration of dispensations of medicine dispensed in Country B based on prescriptions administered in Country A. The service notifies a NCP-A on actions taken by a NCP-B regarding the dispensation of medicine described in retrieved ePrescriptions. There are no obligations for the PN's to take action on the

messages passed through the DispensationService. This service uses the XDR Service implementation.

2.4.1.4 Original Clinical Document Service

This service is used to retrieve original clinical documents; given a valid patient identifier and that the patient has given consent for the distribution of this information. This service uses the XCA Service implementation.

2.4.2 FHIR Service Implementations

The FHIR services are implemented against FHIR R4 (https://hl7.org/fhir/r4/). Migration to R5 (version used by most of the member states) would require updates to the HAPI dependency versions and all resource profiles, Implementation Guide packages, and national connector contracts, and is planned as a future change proposal.

2.4.2.1 EU Lab Report Service

This service is used to retrieve lab result reports. This service allows to search for all lab result reports associated to a patient and then get all the details to a single report. This service uses the DocumentReference and Bundle FHIR resource implementation.

2.4.2.2 Hospital Discharge Report Service

This service is used to retrieve hospital discharge result reports. This service allows to search for all discharge result reports associated to a patient and then get all the details to a single report. This service uses the DocumentReference and Bundle FHIR resource implementation.
Alternatively, it is also possible to push a hospital discharge report from the coutry providing the care to the patient's country of affiliation

2.4.2.3 Medical Imaging Report Study Service

This service is used to retrieve Meidcal imaging reports. This service allows to search for all medical imaging reports associated to a patient and then get all the details to a single report. With the response from the service is then possible to connect to an imaging study repository server (like orthanc) to retrieve the images associated with the patient's report.

2.4.2.4 Vaccination History Retrieval Service

This service is used to retrieve a patient's Vaccination History report. This service uses the DocumentReference and Bundle FHIR resource implementation.

2.4.3 Common Service Implementations

This section describes services that exist in both implementations (NCP and FHIR).

2.4.3.1 Identification Service

The Identification Service is used for obtaining or creating a unique identifier of the patient, which has to be used for all subsequent calls for the medical data of this patient, based on traits supplied by the HP in Country B. Identity management for patients is the concern of Country A and the identification of patients is performed by the National Infrastructure. Based on the information gathered from the PN's use of identity management, it seems that almost every country has implemented integration to their own Identity Providers.
The OpenNCP implementation should provide integration points for obtaining the patient identity information provided by the National Infrastructures. The service will communicate through the XCPD Service implementation. On FHIR we will use the */patient* endpoint to identify the patient we are gathering information from.

2.4.3.2 EUDI Wallet-Based Patient Identification

The Identification Service described in section 2.4.3.1 relies on the HP manually supplying patient traits at the NCP-B Front-End Portal, which are then forwarded to the national infrastructure via the XCPD service. CP090 introduces an optional alternative: the HP may trigger a EUDI-Wallet presentation request, causing the patient to present their member-state-issued Person Identification Data (PID) credential directly from their EUDI-Wallet app. The verified PID claims are mapped to XCPD traits automatically, eliminating manual data entry and transcription errors. This feature is implemented entirely within the NCP-B boundary as a new Maven module openncp-eudi-connector. No changes are required to NCP-A or to any existing OpenNCP module.

The openncp-eudi-connector integrates with the existing client connector through a dedicated EudiIdentificationProvider interface that the client connector calls before building the XCPD findEntityByTraits() request. The integration point is implemented as a Spring bean that can be injected into the existing PatientIdentificationService class.

2.4.3.3 Patient Service

This service is used to retrieve patient summaries; given a valid patient identifier and that the patient has given consent for the distribution of this information.
On NCP this service uses the XCA Service implementation.

2.4.3.4 Anomaly Detection System

This subcomponent is responsible to detect anomalies during the exchanges between NCP-A and NCP-B and allows an administrator to take action.
The possible anomalies are:

1. An excessive number of requests received by NCP-A from any NCP-B within a specified time interval.
2. An excessive number of requests received by NCP-A from the same NCP-B within a specified time interval.
3. An excessive nuber of requests received by NCP-A from the same NCP-B within a specified time interval, and concerning the same patient.

Anomalies have to be checked for XCPD, XCA and XDR services. The Anomaly detection for general traffic rules is warning-only and does not impact workflow. For emergency access abuse specifically, countries are responsible for configuring enforcement at the national policy level.

Component and Dependency Graph OpenNCP Service Consumer

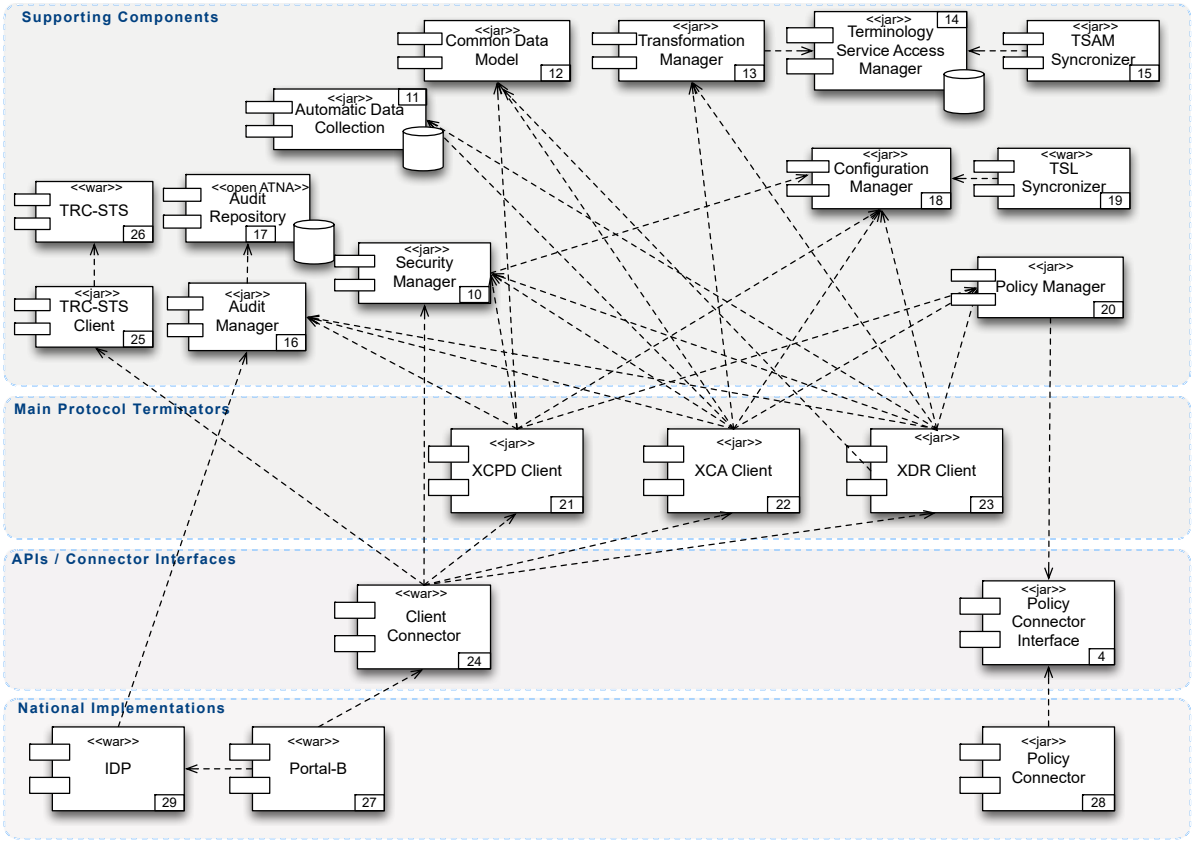
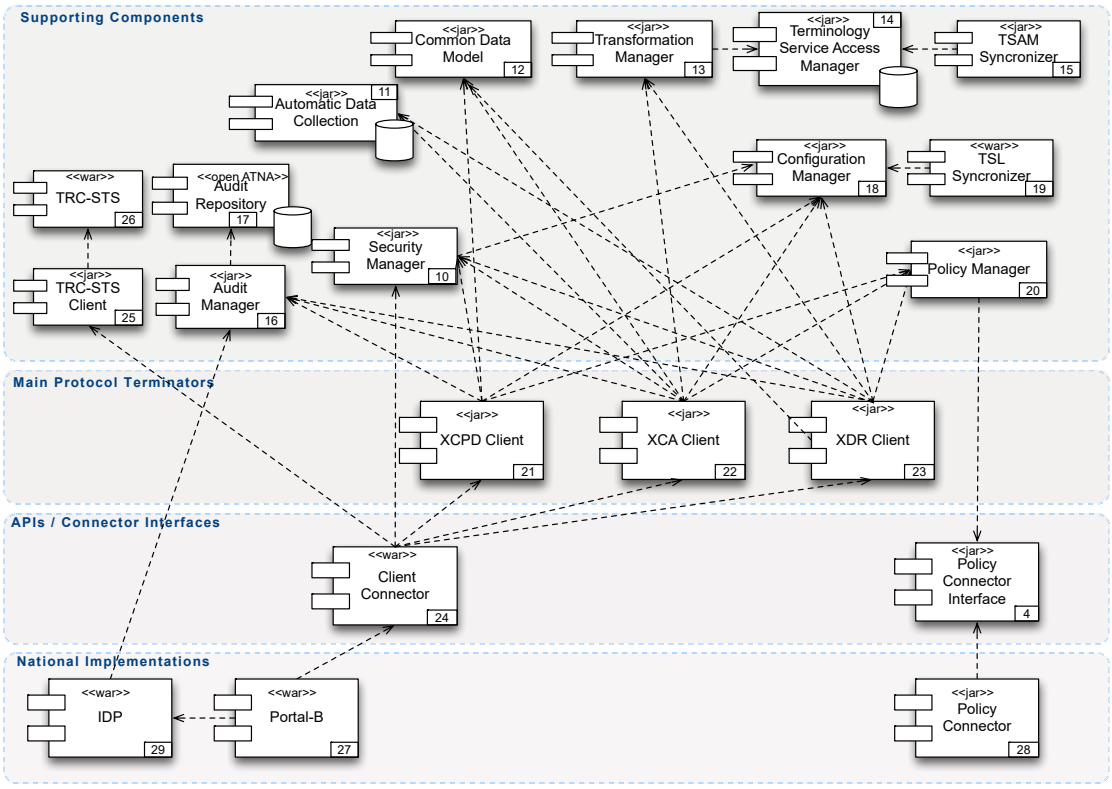


Figure 5: NCP-B architecture describing deployable components and the dependencies between them

Component and Dependency Graph OpenNCP Service Consumer



2.4.4 NCP-B – Client Connector

The preferred approach is to expose a client API supporting the eHealth DSI workflows. The service should implement an open protocol and be designed as a simple stateless API. Thereby the API should be usable in any web portal solution.

2.4.5 Supporting Components

2.4.5.1 Consent Manager

We recommend that the effort is kept to a minimum for the OpenNCP implementation and put on each PN to provide their specific implementation of choice. The OpenNCP implementation provides the PN's with an integration point (figure 3 & 5).

2.4.5.2 Audit Manager

The Audit Manager/Service including client stubs is open source, stable and should be completely re-usable. The OpenNCP implementation should adapt these as external dependencies and provide for a way to plug them in to the NCP Infrastructure.

2.4.5.3 Configuration Manager

The current implementation of the configuration manager keeps the scope of the delivery to a minimal viable solution. Improvement requirements have already been identified (a placeholder for a change proposal is already set) but for maturity and realibility sake (demanded for routine operations) will be included in later releases.

2.4.5.4 Translations and mappings services

The translations and mappings services API handles

- The mapping of information gathered from the NI as a Friendly-A document to a common format known as the eHealth DSI pivot document
- The retrieval of all language codes available for a given jurisdictional domain
- The translation of a Pivot document to a language available for this jurisdictional domain. The eHealth DSI pivot document is an intermediate format that is meant to build a bridge between the national information formats and the HL7 CDA format as well as to serve audit and log purposes.
- The translation of a full value set to a language available for this jurisdictional domain.

2.4.5.5 eHealth DSI Policy Manager

The aim of the OpenNCP implementation is to be compliant with the eHealth DSI specifications regarding the use of SAML. The first implementation approach is to adopt the SRDC NCP solution. In the SRDC solution, the country level policy enforcement is rudimentary (i.e. hard coded based on permission attributes of the requestor) and does not implement the full XACML data flow described in [Security Services Specification]. The OpenNCP implementation provides an integration point in the form of a standard Java interface that PN's can implement (figure 3, 4 & 5). Permission validation for OrCD Lab Results, Medical Imaging, and Hospital Discharge Report services currently applies a temporary fallback to Patient Summary permission checks pending dedicated validators. Countries implementing the NCP are responsible for extending the Policy Manager with service-specific permission enforcement at the national connector level.

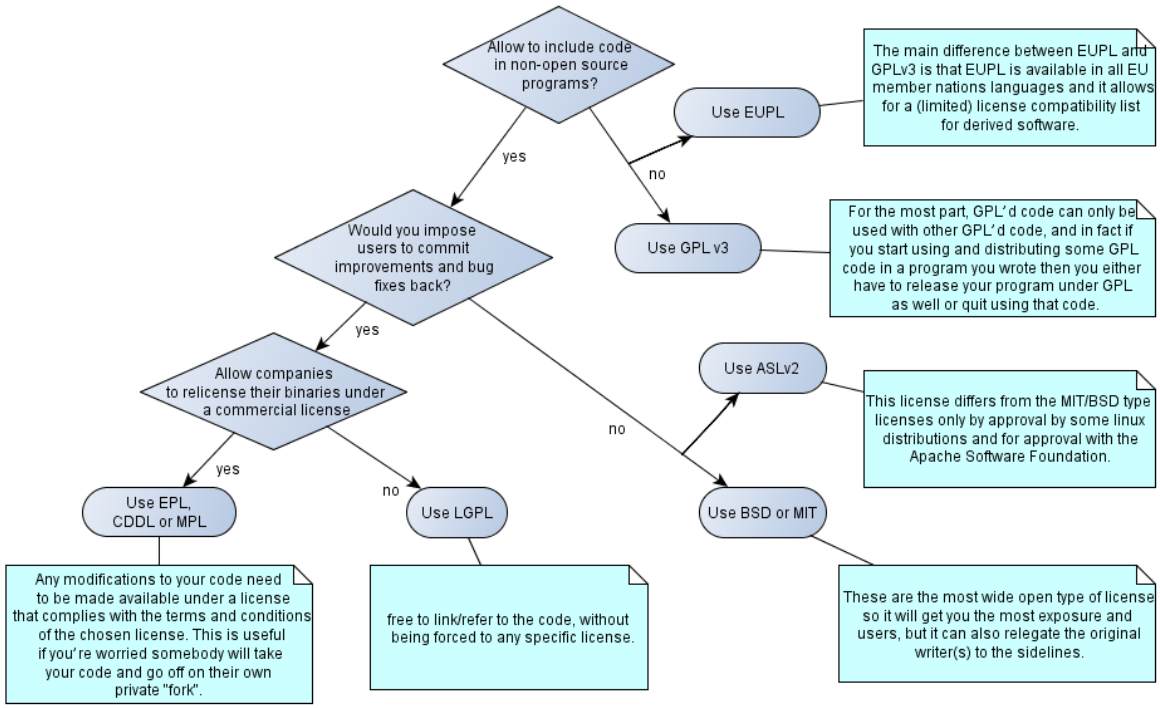
2.4.5.6 eHealth DSI Security Manager

The current security module handles security manager SAML assertion signing/verification, XML DSig, X.509 certificate validation, OCSP, and JWT wrapping for FHIR services.

2.5 License Terms of OpenNCP Compoonents

2.5.1 Introduction

The diagram in figure 6 lists some of the most popular open source licenses and their consequences to the software distributed under these licenses.



Licenses	Can be used in close source	Programs that includes this code can be sold for profit	Source to modifications and bug fixes must be released
GPLv3	Not in derived works	Not for license fees	Yes, if distributed
BSD/MIT	Yes	Yes	No
LGPL	Yes	Yes	Yes, if distributed
EPL/CDDL/MPL	Yes	Yes	Yes, but only the modifications
ASLv2	Yes	Yes	No
EUPL	not in derived works unless released under a permissive license	not unless released under a permissive license	Yes, if distributed under a license that mandates so

Source:
http://www.opensource.org/licenses/alphabetical

Acronyms:
ASLv2: Apache Software License v2
BSD: Berkeley Software License (revised version)
CDDL: Common Development and Distribution License
EPL: Eclipse Public License
EUPL: European Union Public License
GPLv3: GNU Public License v3
LGPL: Lesser GNU Public License
MIT: MIT License
MPL: Mozilla Public License

Notes to the matrix on the left:
The matrix is a very simplified comparison matrix, and there are subtle differences to the licenses, which will be apparent upon closer scrutiny

Disclaimer:
this is just intended for informal, purposes. It is not a substitute for advice from a qualified legal professional.

Figure 6: OSS license decision diagram them

2.5.2 Licensing of the eHealth DSI OS components

All software components which are newly developed by eHealth DSI will be exclusively licensed under the ASL v.2 scheme. Developers of software components are given the responsibility to check that no other licensing rights pertain to any elements embedded or related to a newly developed component, particularly but not limited to libraries.

The OpenNCP community is responsible in co-operation with the eHealth DSI PN's, to ensure that the licenses of all open source components used in the project are compatible with the purpose and scope of the project.

2.5.3 Use and consequence of the licenses

This project is developed under the ASLv2 (eHealth DSI developed components) and GPLv3 licenses (SRDC contributed elements). As the GPLv3 is a strong copyleft license, some care has to be taken when linking between the different components of this project. In general, software libraries developed under ASLv2 can be used by GPLv3 licensed applications while both retain their license terms (see the Apache text on GPL compatibility at http://www.apache.org/licenses/GPL-compatibility.html). However, the other way around, the GPLv3 license will invoke the clauses set forth in section 10 of its license text:

> 10. Automatic Licensing of Downstream Recipients.
> Each time you convey a covered work, the recipient automatically receives a license from the original licensors, to run, modify and propagate that work, subject to this License. You are not responsible for enforcing compliance by third parties with this License.

Specifically, when using any of the components that are licensed under the GPLv3, potential clients of these applications or libraries MUST take special care to observe its license.
To sum it up: In general there are no problems with linking within the project components, as the components licensed under ASLv2 can be used by the GPLed part without license consequences. The issue arises the other way around, when ASLv2 licensed code links against GPLed code. Thus, special care has to be taken when thirdparty libraries are used by the ASLv2 licensed components.
The individual miniprojects are responsible for providing a list of the licenses used by their respective components. This list resides in a file named README located at the root of the project, alongside its *pom.xml* file.
Using GPL licensed software from closed source software has the consequence that such software is subject to the terms stated in the GPL license, because closed software licensed code cannot link directly to GPL licensed code without having to disclose the source code in some form (either through a public repository or through requests of disclosure). This in practice means that it is not possible to directly use GPL licensed software from closed source software without disclosing the closed source software.

2.5.4 Open source software

The software elements are licensed under the GPLv3 scheme.
In Section 2.10, the list of OpenNCP components is provided, indicating which of them are affected by the GPLv3 scheme.

2.6 eHealth DSI Technical Community (OpenNCP Community)

The OpenNCP implementation is steered, managed and developed by the eHealth DSI Technical Community.

The eHealth DSI Solution Provider has a permanent representation in the 3 levels of community governance (Steering Committee, Technical Committee and Development Team).

For more information, please visit eHealth DSI Technical Community knowledge base.